

Project 3: Monocular Metric Depth Estimation & Video Consistency

AIAA 3201—Introduction to Computer Vision

Term Project Instruction

Spring 2027



1 Overview

Core Task: Given a single RGB image or an ordinary monocular video, students are required to estimate dense depth, recover a meaningful metric scale, and maintain stable predictions over time. The project connects classical stereo geometry with modern depth foundation models and video temporal modeling. The final system should support an application such as point-cloud visualization, distance measurement, depth-aware effects, or obstacle warning.

Key Concepts:

- **Projective Geometry:** Disparity, focal length, camera intrinsics, and metric scale.
- **Monocular Depth:** Relative depth versus zero-shot metric depth.
- **Foundation Models:** Large-scale pretraining and cross-dataset generalization.
- **Video Consistency:** Temporal propagation, optical flow, scale drift, and occlusion.

Group Size: 2 students per group.

2 Submission Requirements

- **PDF Report (Mandatory):**

- **Format:** Use the CVPR LaTeX Template. The length must be **6–8 pages** (excluding references).
- **ArXiv Upload:** You are required to upload your final report to **arXiv**. Remember to change the “REVIEW version” to “CAMERA-READY version” in the template. Please clearly indicate your arXiv ID on the first page of the Canvas submission.
- **Content Requirements:**
 1. **Abstract & Introduction:** Background, motivation, problem definition, and solution overview. **Explicitly include your public GitHub repository link at the end of the abstract.**
 2. **Related Work:** Review and cite at least all papers listed in the Recommended Reading List (Section 4). Expanding beyond this list is encouraged.
 3. **Method:** Explain the technical roadmap **in your own words**. Use high-quality pipeline diagrams and visual examples to support the explanation.
 4. **Experiments:** Present quantitative results in tables, qualitative comparisons in figures, ablation studies, and categorized failure cases.
 5. **Conclusion:** Summarize the findings, limitations, and specific directions for future work.

- **Code (Mandatory):** Upload the code to a public **GitHub repository** with a clear README.md. Include dependencies, data preparation, model weights, training and evaluation commands, and representative visual results. Do not submit raw code files to Canvas.
- **Depth Demo & Video (Mandatory):** Submit processed videos for all mandatory clips and a runnable or recorded application demo. The demo must show the RGB input, predicted depth, and one application output such as a point cloud, click-to-measure interface, 2.5D parallax, or depth-aware rendering. Pack the outputs into `depth_results.zip`.

Important Tips for Success

- **Method Flexibility:** You may follow the suggested roadmap or design another technically sound method. Strict adherence to the Implementation Roadmap is not required if the core task and evaluation protocol are fulfilled.
- **Completeness:** All three parts are mandatory. If Part 3 does not improve the main metric, analyze why and provide controlled ablations and failure cases. Ignoring Part 3 will result in an “Incomplete” grade.
- **Visuals:** Clean pipeline diagrams, prediction overlays, zoomed comparisons, and failure visualizations are essential in a Computer Vision course and will substantially affect the presentation score.
- **Dataset Scope:** Completing the downloadable Middlebury Stereo v3 and NYU Depth V2 benchmarks ensures a passing grade. No additional public dataset is required; high-scoring projects should focus on stronger analysis, temporal tests on self-recorded clips, and a convincing application demo.
- **Fair Comparison:** Use identical data splits, preprocessing, input resolution, and evaluation code when comparing methods. Report runtime, peak GPU memory, and all test-time refinements.

- **Citations:** Cite all datasets, pretrained models, external repositories, and borrowed implementation components correctly.

3 Guide to Start

To help you get started efficiently, we recommend the following workflow:

1. **Step 1: Literature Review.** Read the papers in the Recommended Reading List (Section 4). Understanding pinhole geometry, relative versus metric depth, monocular scale ambiguity, optical-flow warping, and depth evaluation protocols is essential before coding.
2. **Step 2: Code Familiarization.** Clone and run the official repositories for Depth Anything V2 and Video Depth Anything. Run their provided demos first, then verify the input/output format and pretrained checkpoint on one sample.
3. **Step 3: Experimentation.** Download the datasets and checkpoints from Section 6.1. Reproduce fixed-image metrics on the specified public split before modifying the temporal pipeline. Save raw floating-point depth maps in addition to colored visualizations.
4. **Step 4: Writing & Submission.** Organize quantitative results, ablations, and failure cases into tables and figures. Write the report with the CVPR template, cite all references, upload the camera-ready manuscript to arXiv, and submit the required files to Canvas.

4 Recommended Reading List

- **Classical Geometry:** Semi-Global Matching [1], KITTI benchmark [2]
- **Monocular Foundations:** DPT [3], Depth Anything [4], Depth Anything V2 [5]
- **Metric Depth:** UniDepth [6], Depth Pro [7], UniDepthV2 [8]
- **Video Depth & Motion:** RAFT [9], Video Depth Anything [10]

5 Implementation Roadmap

5.1 Part 1: The Baseline – “Geometry & Frame-Wise Depth”

Goal: Understand the geometric meaning of depth and establish frame-wise baselines before using temporal models.

1. Classical Stereo: Disparity to Depth

- **Step 1: Rectified Stereo.** Use OpenCV StereoBM or StereoSGBM on the downloaded Middlebury rectified pairs. Visualize the disparity, invalid regions, and left-right consistency.
- **Step 2: Metric Conversion.** Convert disparity d to depth $z = fB/d$ using focal length f and baseline B . Study the sensitivity to disparity errors at different distances.

2. Monocular Frame Baseline

- **Relative Depth:** Run MiDaS/DPT or Depth Anything V2-Small independently on each frame.
- **Scale Alignment:** For benchmark evaluation only, apply the official median or scale-and-shift alignment where required. Do not call aligned relative depth “metric depth.”

- **Temporal Diagnosis:** Measure flicker on a static-camera clip and scale drift on a moving-camera clip. Visualize frame-to-frame depth changes.

Expected Result: Frame-wise models should recover convincing scene structure but may have incorrect metric scale, unstable boundaries, and visible temporal flicker.

5.2 Part 2: SOTA Reproduction – “Metric & Temporally Consistent Depth”

Goal: Reproduce recent open models for zero-shot metric depth and long-video consistent depth without training a foundation model from scratch.

1. Single-Image Metric Depth: UniDepthV2 [8] or Depth Pro [7]

- **Requirement:** Choose one method with released code and weights. Predict metric depth without test-set scale alignment and report the official metric-depth protocol.
- **Key Learning Point:** Analyze the role of camera intrinsics or focal-length prediction and compare boundary quality, absolute scale, and domain generalization.

2. Video Depth: Video Depth Anything [10]

- **System:** Use the official Small or Base checkpoint to infer temporally consistent relative or metric depth for short self-recorded clips. The repository provides batch and streaming inference; no additional public video dataset is required.
- **Required Comparison:** Compare frame-wise Depth Anything V2 with Video Depth Anything using identical resolution, visualization range, and evaluation frames.
- **Resource Rule:** Prefer the Small model and FP16. Do not use the Large model if it exceeds the group’s assigned GPU memory.

Expected Result: The video model should reduce flicker and scale drift, while thin structures, specular surfaces, rapid motion, and occlusions may remain difficult.

5.3 Part 3: Exploration – “Calibration, Consistency & Application”

Goal: Identify one measurable limitation of the reproduced system and implement a lightweight improvement with controlled ablations.

- **Direction A (Flow-Guided Fusion):** Warp the previous depth map with RAFT or Farnebäck flow, reject occlusions with forward-backward consistency, and fuse it with the current prediction using confidence weights.
- **Direction B (Metric Scale Calibration):** Use a known-size object, camera height, ground plane, or sparse depth samples to estimate scale. Study robustness when the calibration cue is noisy.
- **Direction C (Edge-Aware Refinement):** Use RGB gradients, semantic boundaries, or segmentation masks to sharpen depth discontinuities without copying texture into depth.
- **Direction D (Uncertainty-Aware Smoothing):** Estimate uncertainty from model confidence, test-time augmentations, or cross-model disagreement. Smooth only uncertain and temporally inconsistent regions.
- **Direction E (Efficient Deployment):** Export a Small model to ONNX/Core ML, apply quantization, or use dynamic resolution. Report speed, memory, and accuracy trade-offs.
- **Direction F (Depth-Aware Application):** Use the depth in a measurable downstream task such as object-distance ranking, AR occlusion, portrait bokeh, or collision warning. Evaluate the application output, not only visual quality.

6 Dataset & Evaluation

6.1 Datasets and Required Downloads

1. **Middlebury Stereo v3 Quarter Resolution (Mandatory):** Download the 30 MB quarter-resolution input ZIP and the 13 MB left-view ground-truth ZIP. Both are direct downloads with no account or request form. Use the 15 training pairs for StereoBM/StereoSGBM, disparity-to-depth conversion, and error analysis; calibration files are included with the inputs. [Direct input ZIP](#) | [Direct ground-truth ZIP](#) | [Official benchmark page](#)
2. **NYU Depth V2 Labeled Dataset (Mandatory):** Download the single 2.77 GB labeled MATLAB file directly; do not download the 428 GB raw collection. No account or request form is required. Use the standard NYUv2 evaluation split and the protocol provided by the Depth Anything V2 metric-depth code; do not train a foundation model from scratch [12]. [Direct labeled dataset file](#) | [Official dataset description](#) | [Depth Anything V2 metric-depth tools](#)
3. **Starter Code and Checkpoints (Required):** Use Small checkpoints when GPU memory is limited. Each official repository links its pretrained weights and inference commands. [Depth Anything V2](#) | [Video Depth Anything](#) | [UniDepthV2](#) | [Depth Pro](#)

6.2 Metrics (Mandatory)

- **Metric Depth Accuracy:** AbsRel, SqRel, RMSE, RMSE-log, and $\delta_1/\delta_2/\delta_3$. State whether predictions use metric scale or test-time alignment.
- **Boundary Quality:** Depth-edge precision/recall or a justified boundary error on selected scenes with clear discontinuities.
- **Temporal Consistency:** Optical-flow-aligned temporal error on valid non-occluded pixels and per-frame median-scale drift.
- **Efficiency:** FPS, latency per frame, peak GPU memory, model size, and input resolution.
- **Qualitative Evaluation:** Show RGB, colored depth with a fixed scale, 3D point cloud or application output, and failure cases on mirrors, glass, sky, motion blur, and thin objects.

Note: A colored depth video with independently normalized frames can hide flicker. Use a fixed visualization range for every method on the same clip.

References

- [1] Hirschmüller, H. “Stereo Processing by Semiglobal Matching and Mutual Information.” TPAMI 2008.
- [2] Geiger, A., et al. “Are We Ready for Autonomous Driving? The KITTI Vision Benchmark Suite.” CVPR 2012.
- [3] Ranftl, R., et al. “Vision Transformers for Dense Prediction.” ICCV 2021.
- [4] Yang, L., et al. “Depth Anything: Unleashing the Power of Large-Scale Unlabeled Data.” CVPR 2024.
- [5] Yang, L., et al. “Depth Anything V2.” NeurIPS 2024.

- [6] Piccinelli, L., et al. “UniDepth: Universal Monocular Metric Depth Estimation.” CVPR 2024.
- [7] Bochkovskii, A., et al. “Depth Pro: Sharp Monocular Metric Depth in Less Than a Second.” ICLR 2025.
- [8] Piccinelli, L., et al. “UniDepthV2: Universal Monocular Metric Depth Estimation Made Simpler.” arXiv:2502.20110, 2025.
- [9] Teed, Z., and Deng, J. “RAFT: Recurrent All-Pairs Field Transforms for Optical Flow.” ECCV 2020.
- [10] Chen, S., et al. “Video Depth Anything: Consistent Depth Estimation for Super-Long Videos.” CVPR 2025.
- [11] Sturm, J., et al. “A Benchmark for the Evaluation of RGB-D SLAM Systems.” IROS 2012.
- [12] Silberman, N., et al. “Indoor Segmentation and Support Inference from RGBD Images.” ECCV 2012.